SSC CGL

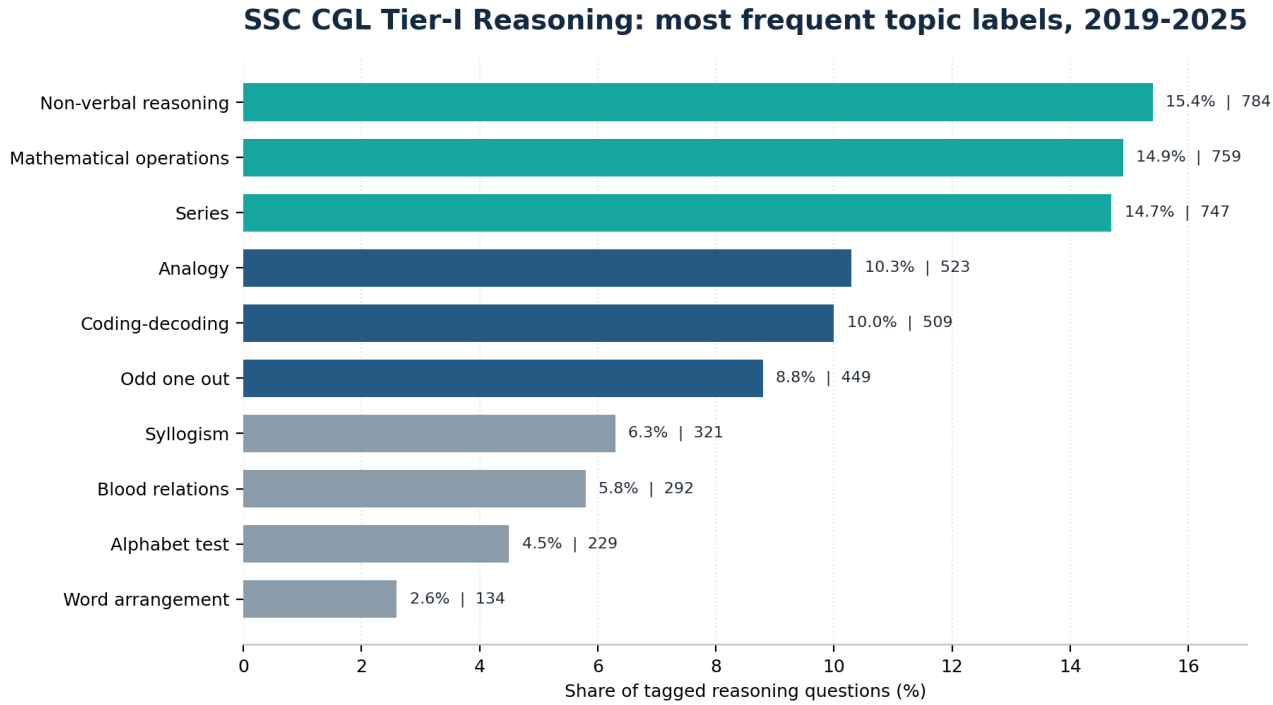
REASONING MASTER SYLLABUS

Tier-I + Tier-II | Priority map | Historical trends | 24-week execution plan

TIER-I	TIER-II	HISTORICAL SIGNAL
25 questions	30 questions	45%
50 marks | 15 min	90 marks | 30 min	Top 3 topic labels

THE CENTRAL STRATEGY

Master repeatable pattern families first, then cover every official syllabus item. Historical frequency tells us where to spend extra practice; it does not guarantee the composition of the next paper.



Source: Comperly Acad analysis of 5,077 topic-tagged questions across 204 papers (2019-2025). Classification is source-specific; not an SSC forecast.

A data-informed view of what has appeared often in one third-party tagged sample of SSC CGL papers (2019-2025).

Built against the SSC CGL 2026 official notification | Version date: 30 October 2026

1. Know the exam before planning the syllabus

Reasoning appears in both Tier-I and Tier-II Paper-I. The official topic lists are closely related, but the number of questions, scoring and time available are different. This changes how the same knowledge must be practised.

Requirement	Tier-I	Tier-II Paper-I
Section name	General Intelligence & Reasoning	Reasoning & General Intelligence
Questions	25	30
Marks	50	90
Reasoning time	15 minutes	30 minutes
Marks per correct answer	+2	+3
Penalty per wrong answer	-0.50	-1
Practical focus	Fast recognition; skip/return decisions within section timer	Stronger consistency and disciplined pace in a longer, higher-value section

What the official syllabus says

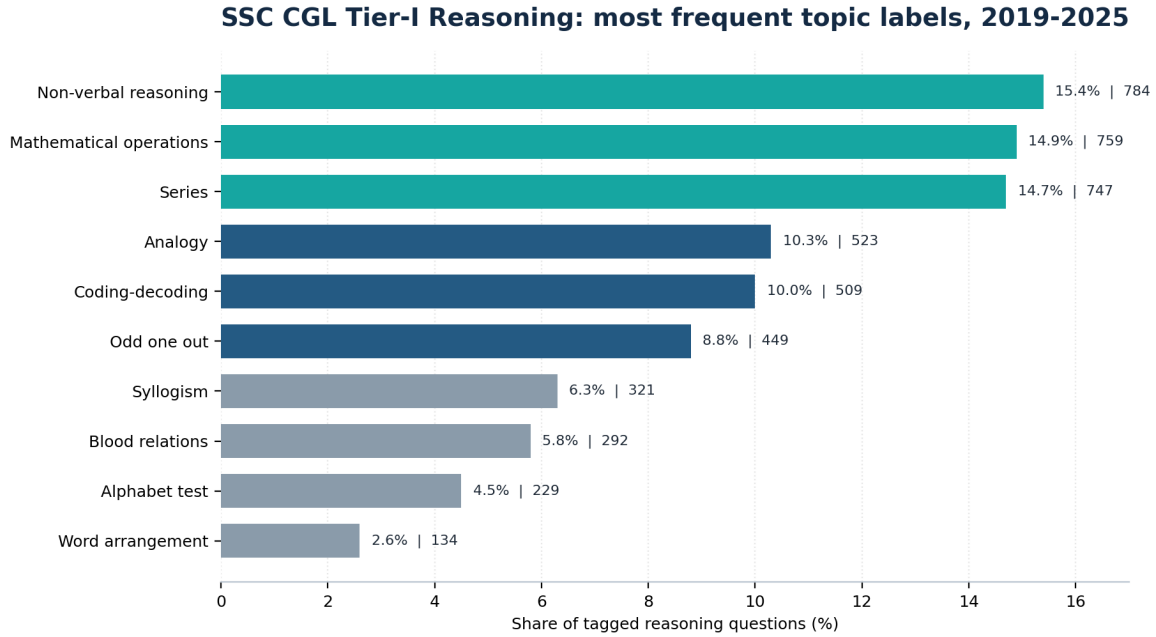
- Both verbal and non-verbal reasoning are included.
- Tier-I explicitly names analogy, classification, semantic/number/figural series, problem solving, word building, coding, numerical and symbolic operations, trends, space orientation/visualization, Venn diagrams, inference, folding/unfolding, indexing, matching, embedded figures, critical thinking, emotional intelligence and social intelligence.
- Tier-II Section-I Part B repeats most core families and says “other sub-topics, if any”; prepare the complete official Tier-I list rather than treating only popular coaching chapters as the whole syllabus.

> **IMPORTANT SCOPE NOTE**
>
> Blood relations, direction sense, ranking, dice, cubes, clock/calendar-style tasks, seating and puzzles are commonly used practice formats across competitive-reasoning preparation. Some fit under broader official labels such as relationship concepts, spatial orientation, problem solving or numerical operations; they are not all separately named in the SSC wording.

Official basis: SSC CGL 2026 notification, paras 13.8, 13.9 and 13.10.1 / 13.11.2. See source list at the end.

2. Historical data: what deserves extra practice?

A third-party analysis reports 5,077 topic-tagged General Intelligence and Reasoning questions across 204 SSC CGL papers from 2019-2025. Its three largest labels are non-verbal reasoning, mathematical operations and series.



Source: Comperly Acad analysis of 5,077 topic-tagged questions across 204 papers (2019-2025). Classification is source-specific; not an SSC forecast.

Historical cluster	Count	Share	Planning implication
Non-verbal reasoning	784	15.4%	Build daily visual recognition drills.
Mathematical operations	759	14.9%	Practise operator substitution and numerical/symbolic transformations.
Series	747	14.7%	Learn a pattern-testing checklist; drill number, letter and figure series.
Analogy	523	10.3%	Compare relation types, not surface similarity.
Coding-decoding	509	10.0%	Practise letter/number shifts, position values, reversal and mixed rules.
Odd one out / classification	449	8.8%	State the common rule for the valid group and test every item.

HOW TO READ THIS CHART

The top three labels sum to 45.0% in this tagged sample. This is historical description, not official weightage or a guarantee for future shifts; do not skip lower-frequency official topics.

Source: Comperly Acad, "SSC CGL Topic-Wise Weightage: What 204 Real Papers From 2019-2025 Show" (published 6 September 2026). Counts depend on tagging choices and dataset coverage.

3. Complete syllabus map - every official family in one view

Use these families as a reusable checklist. They preserve the exact named scope of the SSC notification, while adding sensible subtopic labels so daily practice is easier to organise.

Family	Officially named topics grouped for study	Priority
A. Pattern relationships	Semantic, symbolic/number and figural analogy; semantic, symbolic/number and figural classification; semantic, number and figural series; trends.	P1
B. Codes and operations	Coding-decoding; small/capital letters and number coding, decoding and classification; numerical operations; symbolic operations.	P1
C. Non-verbal / spatial	Space orientation; space visualization; figural pattern folding/completion; punched-hole/pattern folding and unfolding; embedded figures; visual memory; observation; discrimination.	P1
D. Inference and formal logic	Venn diagrams; drawing inferences; statement-conclusion; syllogistic reasoning; critical thinking; analysis; judgment; decision making; problem solving.	P1-P2
E. Relationships and applied reasoning	Relationship concepts; arithmetical reasoning; problem solving; emotional intelligence; social intelligence.	P2-P3
F. Word and matching tasks	Word building; indexing; address matching; date and city matching; classification of centre codes/roll numbers.	P2-P3

What not to do

- Do not study only “analogy, series, coding and blood relation” and assume that the syllabus is complete.
- Do not memorise a fixed number of questions per topic from a coaching chart. SSC does not publish official topic-wise weightage.
- Do not equate “less frequently tagged” with “not examinable”. Every officially listed family belongs in your coverage audit.

> **MASTER RULE**
>
> The syllabus is complete only when you have at least one learning note, one practice set and one PYQ/review record for each named family.

4. Pattern engine: analogy, classification, series and codes

These questions reward structured observation. Your job is not to guess quickly; it is to identify a rule, verify it across all relevant elements, and reject attractive but incomplete patterns.

4A. Analogy and classification

- ☐ Semantic analogy: function, part-whole, cause-effect, tool-user, material-product, degree, synonym/antonym and category relationships.
- ☐ Symbolic/number analogy: arithmetic transformation, place value, squares/cubes, multiples, digit sums and combined operations.
- ☐ Figural analogy: rotation, reflection, movement, count, shading, orientation, insertion/removal and position change.
- ☐ Classification / odd one out: common property, letter positions, shape features, mathematical grouping, semantic category and exception detection.

4B. Series and trends

Series family	Test systematically
Number series	Differences; second differences; ratios; alternating tracks; squares/cubes; primes; digit operations; mixed patterns.
Semantic / letter series	Alphabet positions; forward/backward gaps; alternating letters; grouped pairs; position transformations; wrap-around rules.
Figural series	Rotation/reflection; moving dot/line; number of elements; fill/shade; addition/deletion; alternating frames.
Trends	Observe how one attribute changes across the options or sequence; state a single rule that works consistently.

4C. Coding-decoding and operations

- ☐ Letter codes: fixed shifts, opposite letters, reversal, alternate positions, position numbers, sums and mixed transformations.
- ☐ Number/symbol codes: digit substitution, numerical sequence, arithmetic encoding, letter-number conversion and table-based mapping.
- ☐ Numerical/symbolic operations: replace the symbols exactly as instructed, then apply precedence carefully; test candidate transformations against every given example.

> **PATTERN CHECKLIST**
>
> Before choosing an option, ask: (1) What changes? (2) What stays fixed? (3) Is the rule consistent for every term? (4) Can any second option also fit? If yes, the rule is not yet strong enough.

5. Non-verbal and spatial reasoning

The historical sample labels non-verbal reasoning as the largest topic group. Practise it visually, not only through written descriptions. Draw, mark, rotate mentally, and compare one feature at a time.

Topic	Subskills to practise	How to check your method
Space visualization	2D/3D transformation, shape position, component rotation and orientation.	Track one distinctive edge or mark before moving the whole shape.
Space orientation	Direction of objects/persons, rotations, relative positions and spatial viewpoints.	Record the starting direction and each turn explicitly.
Figure series / trends	Rotation, reflection, position shifts, element count, shading and alternating patterns.	Describe the change from frame to frame in one sentence.
Embedded figures	Find a smaller shape inside a more complex design.	Ignore size and surrounding lines; scan angles, intersections and orientation.
Paper folding / punched holes	Folding, symmetry, hole reflection and unfolding positions.	Reverse each fold in the opposite order.
Pattern folding / completion	Choose a completed or unfolded shape that satisfies the original pattern.	Check corners, edge matching, orientation and symmetry.
Figural classification	Compare shape family, number of parts, angles, enclosed regions and direction.	Find a feature shared by three options and missing from the fourth.
Visual memory / observation / discrimination	Spot differences, remember arrangements and identify small changes.	Compare left-to-right and count features rather than relying on a glance.

A three-pass visual solving method

Pass	Action	Time discipline
1. Scan	Count components, note rotation, shading, direction and position.	Do not calculate before you know the feature.
2. Hypothesis	Pick one likely transformation or invariant.	Test it on at least two transitions/options.
3. Verify	Check the full figure/sequence for contradictions.	If the visual rule stays ambiguous, mark and move on.

VISUAL PRACTICE ROUTINE

Complete a small set of non-verbal questions most study days. Improvement comes from recognising repeatable visual operations - not from memorising the answer to a particular diagram.

6. Inference, relations and problem-solving

6A. Inference and formal reasoning

- ☐ Venn diagrams: set relationships, overlap, inclusion, exclusion and translating statements into regions.
- ☐ Syllogistic reasoning: universal/particular statements, definite versus possible conclusions, diagram-first verification.
- ☐ Drawing inferences / statement-conclusion: accept only what follows from the premises; do not import real-world assumptions.
- ☐ Critical thinking, analysis, judgment and decision-making: distinguish facts from assumptions, identify what follows, compare alternatives against stated criteria.
- ☐ Problem solving: multi-step conditions, arrangement and matching grids, information sorting and constraint tracking.

6B. Relationship and applied reasoning

- ☐ Relationship concepts: family-tree questions, coded relationships and multi-person links.
- ☐ Arithmetical reasoning: age relationships, numerical comparisons, operation-based word problems and quantitative conditions.
- ☐ Word building: form valid words, rearrange letters, count positions and check whether the available letters support a candidate.
- ☐ Indexing, address matching, date/city matching, centre-code and roll-number classification: compare fields systematically and avoid positional slips.
- ☐ Emotional and social intelligence: judgement in interpersonal contexts and selecting the most appropriate response supported by the scenario.

6C. Common supplementary practice formats

Format	How to handle it
Direction / distance	Draw a small route and update coordinates/direction after each movement.
Order / ranking	Separate rank from position; check whether the question asks from top, bottom, left or right.
Blood relations	Draw a family tree with one symbol per person and label gender/relationship only when stated.
Dice / cubes	Track opposite faces from separate views; distinguish rotation from reflection.
Calendar / clock	Use only when supported by the exact prompt; write the reference point and each interval clearly.
Seating / floor puzzles	Make a table, place fixed constraints first, and reject any arrangement that violates one condition.

> **OFFICIAL VS STUDY LABELS**
>
> Some popular subtopics are named separately in coaching books but are grouped under broader official concepts. Learn them as useful practice formats, while using the Commission’s wording for your syllabus-completion audit.

7. Tier-I and Tier-II: coverage audit

Tier-II's Reasoning and General Intelligence syllabus is very close to Tier-I. Do not create two completely separate syllabuses. Build one complete coverage map, then raise difficulty, accuracy and pace for Tier-II.

Coverage family	Tier-I notification	Tier-II Paper-I Part B	Study decision
Analogy	Semantic, symbolic/number, figural	Semantic, symbolic/number, figural	Common core; practise multi-step patterns.
Classification	Semantic, symbolic/number, figural	Semantic, symbolic/number, figural	Common core; explain the rule.
Series & trends	Semantic, number, figural; non-verbal series; trends	Semantic, number, figural series; trends	Common core; add mixed and less-obvious rules.
Coding & operations	Coding/decoding; letter/number formats; numerical/symbolic operations	Coding/decoding; numerical operations; symbolic operations	Common core; reduce translation errors.
Visual / spatial	Orientation, visualization, folding, punching, completion, embedded figures	Orientation, folding, completion, embedded figures, figural series	Common core; more timed visual sets.
Logic / inference	Venn, drawing inferences, statement-conclusion, syllogism, critical thinking	Venn, drawing inferences, critical thinking	Keep formal logic practice continuous.
Other listed families	Problem solving, word building, indexing, address/date/city matching, centre code/roll number, emotional/social intelligence	Problem solving, word building, emotional/social intelligence; other sub-topics if any	Cover every Tier-I named task; retain matching and indexing practice.

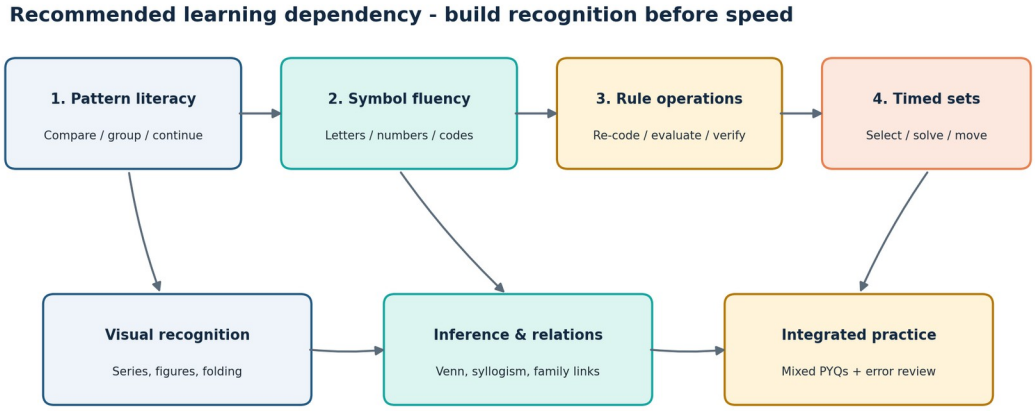
The performance upgrade

Dimension	Tier-I benchmark	Tier-II training direction
Speed	Work toward a full 25-question set in 15 minutes.	Work toward a full 30-question set in 30 minutes.
Accuracy	Build speed only after the rule is stable.	Prioritise precision: wrong answer penalty is 1 mark under the 2026 notice.
Question selection	Identify quick patterns first; do not let one visual puzzle consume the section.	Protect time by limiting prolonged uncertainty; return only if the section timer permits.
Review	Track concept, reading, observation, and calculation errors.	Add time-to-solve and decision-quality analysis to each mock.

Official pattern: SSC CGL 2026 notification, paras 13.8-13.9 and syllabus paras 13.10.1 / 13.11.2.

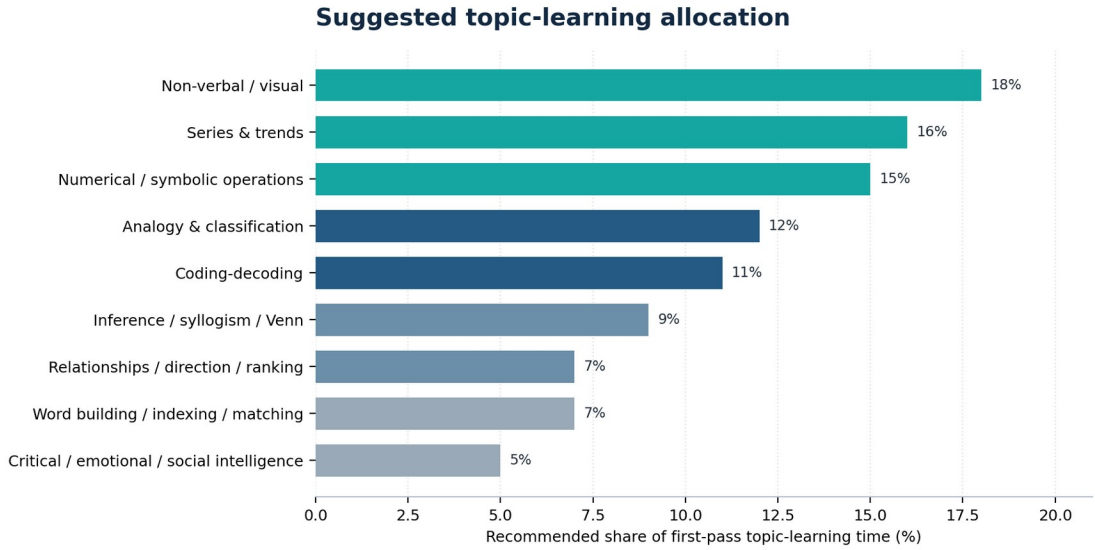
8. Smart learning order - prerequisite first, frequency second

Textbooks often follow a chapter order. This plan follows a learning order: first teach the brain to notice patterns, then give it symbol tools, then move into visual and formal inference tasks, and finally mix everything under timed conditions.



Important: this is a study sequence, not an official SSC question order. Rotate all families into weekly mixed practice.

Suggested allocation of first-pass topic-learning time



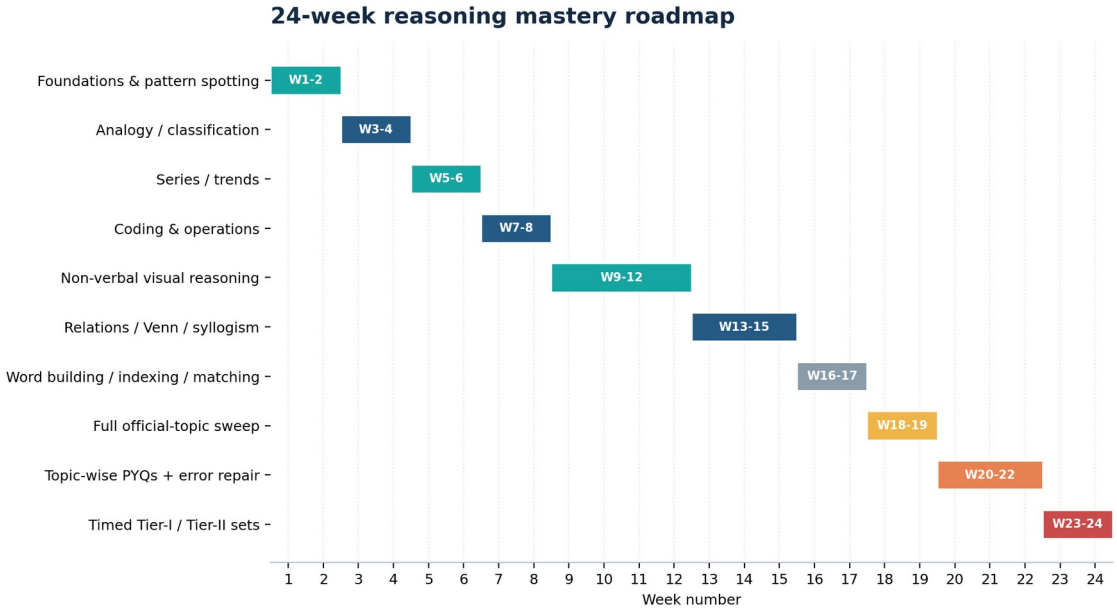
Planning recommendation, not official weightage. Keep PYQ practice, timed sets, and revision running across all topics.

HOW TO USE THE ALLOCATION

Treat these percentages as a starting budget for learning new concepts, not official weightage. After a diagnostic, move time from your strongest families into weak families. Reserve separate time each week for PYQs, mixed sets and error correction.

9. The 24-week execution roadmap

The plan assumes a steady first pass with weekly review. If a family fails the mastery gate, repeat or compress later review blocks instead of moving on with a weak foundation.



Flexible plan. Revisit earlier topics every week; do not postpone all PYQs until the end.

Weeks	Focus	Expected deliverable
1-2	Pattern literacy, visual observation, simple number/letter transformations	Baseline diagnostic; error log set up.
3-4	Analogy and semantic/symbolic/figural classification	Chapter notes + 2 fresh practice sets per family.
5-6	Number, letter, semantic and figural series; trends	A series-pattern checklist; timed mini sets.
7-8	Coding-decoding; numerical and symbolic operations	Transformation rules and verification routine.
9-12	Non-verbal reasoning, space visualization/orientation, embedded figures, folding/punching	Regular image-based drills; error categories separated.
13-15	Venn, syllogism, inference, relationship concepts, problem solving	Diagram-first method; formal conclusion review.
16-17	Word building, indexing, matching, code/roll classification, EI/SI	Complete low-frequency official coverage.
18-19	Second pass of full official list + weak-family remediation	Every official topic has at least one practice record.
20-22	Chapter-wise and mixed PYQs; topic error repair	Solve unseen questions from authentic papers.
23-24	Timed Tier-I and Tier-II sections; post-mock diagnosis	Target speed, accuracy, selection and stability.

10. Daily training system and mastery gates

A 40-minute reasoning block

Time	Activity	Output
5 min	Recall: revise yesterday's patterns and one error-log item.	Warm-up, no passive rereading.
12 min	Learn one concept / review one lesson segment.	A compact rule note plus one solved example.
13 min	Solve fresh questions without viewing the solution.	Accuracy record; mark uncertain items.
7 min	Attempt authentic PYQs from that family.	Learn SSC wording and option design.
3 min	Record error type and next-review date.	A concrete action for tomorrow.

Mastery gates - move forward only when performance supports it

Gate	Suggested benchmark	If not met
Concept clarity	At least 18/20 basic questions correct on a fresh set.	Return to the rule; explain why each distractor fails.
Family mastery	At least 85-90% accuracy on a new mixed set, not a memorised worksheet.	Separate conceptual errors from observation/reading errors.
Tier-I execution	Progress toward 23-25 correct out of 25 in 15 minutes.	Reduce slow question types; drill recognition and skipping decisions.
Tier-II execution	Progress toward 27+ correct out of 30 in 30 minutes while protecting accuracy.	Track time spent per question and rework costly errors.
Retention	Re-solve after 1 day, 1 week and 3 weeks.	If you cannot reproduce the method, the topic is not stable yet.

Four error labels

Code	Meaning	Correction routine
C - Concept	Did not know which rule applied.	Relearn one rule; solve 3 examples with changed surface details.
O - Observation	Missed a visual feature, symbol or condition.	Mark every relevant feature and check the complete stem/options.
R - Reasoning	Rule was incomplete or conclusion went beyond premises.	Write the logic chain; test every option against all conditions.
T - Time	Correct route took too long or selection was poor.	Compare a standard method with a shortcut; practise a timed set.

RANK-LEVEL PRACTICE PRINCIPLE

A high score depends not only on knowing a method but on recognising when to use it, checking it correctly, and abandoning a slow question at the right time. No syllabus can guarantee a particular All-India rank.

11. Resource stack: one course, one PYQ source, one error log

Keep your resources small and role-based. The best resource is one you complete and use for repeated retrieval, not one you collect and leave unfinished.

Role	Resource	How to use it
Official scope	SSC CGL 2026 notification PDF	Use for exam scheme and exact Tier-I / Tier-II syllabus wording.
Main concept course	Adda247 SSC CGL full Reasoning playlist	Follow topic sequence from basics through questions; do not watch several full courses in parallel.
Practice alternative	Adda247 topic-wise practice playlist	Use after concept learning for additional question sets.
Recent focused practice	Vikramjeet Sir / SelectionWay SSC Reasoning Rank Booster	Use for rapid review and timed question-solving after learning foundations.
Free PYQ archive	ExamSIDE: General Intelligence & Reasoning	Use chapter-wise papers and solutions; select SSC CGL Tier-I papers and note year/shift.
Question bank	UnlockIAS: topic-wise SSC CGL Reasoning PYQs	Use to increase volume; verify questionable transcriptions with a second source or official response sheet.
Supplementary sets	PYQ Vault Series / Coding-Decoding	Use for focused practice in high-frequency families; keep a record of wrong-answer patterns.

Clickable resource directory

- Official SSC CGL 2026 notification PDF
- Adda247 complete Reasoning playlist
- Adda247 Reasoning practice questions playlist
- Vikramjeet Sir - Reasoning Rank Booster
- ExamSIDE chapter-wise Tier-I Reasoning PYQs
- UnlockIAS topic-wise Reasoning PYQs
- PYQ Vault: Series
- PYQ Vault: Coding-Decoding
- ExamGOAL chapter-wise Reasoning PDFs
- Historical topic-count analysis

RESOURCE DISCIPLINE

Choose one primary teacher for concept sequence. Use the second source only when a concept is unclear or for a different question style. Your PYQ workbook and error log should be the centre of the plan.

12. Printable master progress tracker

Mark each family only when you have learnt the concept, solved practice questions, attempted PYQs and revisited mistakes. Add a date or accuracy score in the final column.

Topic family	Concepts	Practice	PYQs	Score / review date
Semantic / word analogy				
Symbolic / number analogy				
Figural analogy				
Semantic classification / odd one out				
Symbolic / number classification				
Figural classification				
Semantic series / trends				
Number series				
Figural / non-verbal series				
Coding-decoding: letters and numbers				
Numerical operations				
Symbolic operations				
Space orientation				
Space visualization				
Venn diagrams				
Inference / statement-conclusion				
Syllogistic reasoning				
Problem solving / analysis / judgment				
Word building				
Visual memory / observation / discrimination				
Punched holes / folding / unfolding				
Figural pattern completion				

12A. Progress tracker (continued)

Topic family	Concepts	Practice	PYQs	Score / review date
Embedded figures				
Indexing / address matching				
Date-city matching / centre code-roll number classification				
Critical thinking				
Emotional intelligence				
Social intelligence				
Extended format practice: relations / direction / ranking / dice / puzzles				
Tier-I timed mixed sets				
Tier-II timed mixed sets				

Mock-test log

Date	Test / source	Correct	Wrong	Accuracy	Main error to fix

READY-TO-MOVE-ON CHECK

Before you declare the module complete, you should be able to name every official topic family, solve a fresh topic-wise set, recognise each common visual/logic format, and perform two full-length timed Reasoning sections with an error review afterwards.

13. Sources, evidence and limitations

This syllabus is a study-design document, not a prediction model. It combines the Commission's formal scope with a third-party historical count and topic-wise PYQ repositories.

Primary source: the official SSC notice

Staff Selection Commission, Combined Graduate Level Examination 2026, official notification PDF. Key sections: 13.8 (Tier-I scheme), 13.9 (Tier-II scheme), 13.10.1 (Tier-I General Intelligence & Reasoning) and 13.11.2 (Tier-II Reasoning and General Intelligence).

Open official SSC CGL 2026 notification

Historical distribution source

Comperly Acad, "SSC CGL Topic-Wise Weightage: What 204 Real Papers From 2019-2025 Show," 6 September 2026. The article reports 5,077 tagged reasoning questions. Its topic tags support comparisons inside that database; the grouping need not match another provider's tagging.

Open historical topic analysis

PYQ repositories and teacher resources

- ExamSIDE chapter-wise General Intelligence & Reasoning
- UnlockIAS topic-wise question archive
- ExamGOAL Reasoning topic PDFs
- Vikramjeet Sir / SelectionWay SSC Reasoning Rank Booster
- Adda247 full Reasoning playlist

Method notes

- SSC publishes the indicative syllabus and exam pattern, but does not prescribe fixed topic-wise reasoning weightage.
- Historical topic counts are descriptive, depend on the provider's sample and tagging decisions, and should be used for prioritisation rather than prediction.
- Popular practice categories may fit under broader official syllabus terms; the official notification is the final authority on scope.
- The 24-week schedule, study-time allocation and mastery gates are recommendations. Adjust them to diagnostic performance and the actual recruitment calendar.

FINAL OPERATING PRINCIPLE

Learn the rule. Solve a fresh set. Practise authentic PYQs. Review the error. Revisit it later. Then repeat under the section timer.